

INT374 : DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester August-January 2025)

TGSPDCL Service Performance & Consumption Analysis



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Discipline of CSE

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Lovely Professional University, Phagwara

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Course Code: INT374

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Associate Professor

CERTIFICATE

This is to certify that Ponamala Hemanth bearing Registration No. 12304744 has successfully completed the course project titled “**TGSPDCL Service Performance & Consumption Analysis Using Power BI**” under my guidance and supervision for the course INT374. The work presented is original and meets the academic requirements.

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Date:20/12/2025

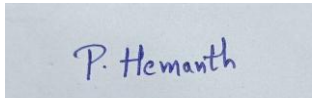
DECLARATION

I, P.Hemanth, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

Date: 20/12/2025

Registration No: 12304744

Signature:



Name: P.Hemanth

ACKNOWLEDGEMENT

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I would also like to thank all faculty members of the School of Computer Science and Engineering for providing the resources and learning environment that enabled me to undertake this research. Lastly, I extend my heartfelt thanks to my family and friends for their moral support and motivation.

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1. INTRODUCTION

The power distribution sector is one of the most data-intensive domains, as it involves continuous monitoring of electricity generation, distribution, consumption, and billing. Electricity distribution companies manage millions of consumers and thousands of service connections, making manual monitoring inefficient and error-prone. Accurate analysis of service performance and consumption data is crucial for ensuring operational efficiency, minimizing losses, and improving customer satisfaction.

The Telangana State Southern Power Distribution Company Limited (TGSPDCL) is responsible for distributing electricity to a large geographical area, covering urban, semi-urban, and rural regions. Due to the scale of operations, TGSPDCL generates extensive datasets related to electricity services, billing records, and unit consumption. Analyzing this data helps the organization identify inefficiencies such as **low billing efficiency, unbilled services, abnormal consumption patterns, and operational gaps** across divisions and sections.

Traditional reporting methods, such as static spreadsheets and manual reports, are often time-consuming and lack flexibility. These methods make it difficult for decision-makers to quickly identify trends or drill down into problem areas. To overcome these limitations, modern organizations are increasingly adopting **Business Intelligence (BI) tools** like Power BI, which provide dynamic visualizations and real-time insights.

Power BI is a powerful data visualization and analytics tool developed by Microsoft. It allows users to connect to multiple data sources, clean and transform data, build interactive dashboards, and share insights with stakeholders. One of the key advantages of Power BI is its ability to simplify complex datasets into easy-to-understand visuals such as charts, tables, and key performance indicators (KPIs). This makes it especially suitable for non-technical users, including managers and policymakers.

This project focuses on analyzing TGSPDCL electricity service and consumption data using **Power BI Desktop**, with a strong emphasis on **Power Query-based data transformation instead of DAX**. By using Power Query, the project demonstrates how data can be cleaned, structured, and calculated efficiently at the backend level. This approach ensures better performance, transparency, and ease of maintenance.

The primary goal of this project is to analyze service performance and billing efficiency across multiple divisions and areas. By comparing total services with billed services and analyzing electricity unit consumption, the project aims to uncover hidden patterns and problem areas that impact revenue and service delivery. Interactive dashboards are developed to allow users to filter, drill down, and explore data from different perspectives.

2. BACKGROUND OF TGSPDCL

The Telangana State Southern Power Distribution Company Limited (TGSPDCL) is a government-owned electricity distribution utility responsible for supplying power to the southern districts of Telangana. It plays a vital role in delivering reliable electricity to domestic, commercial, agricultural, and industrial consumers.

TGSPDCL manages a vast distribution network that includes multiple circles, divisions, subdivisions, and sections. Each level of this structure handles thousands of electricity service connections on a daily basis. Due to increasing population growth, urbanization, and industrial development, the demand for electricity has risen significantly in recent years. This growth has resulted in the generation of large volumes of operational and consumption data. Monitoring service performance and billing efficiency manually has become increasingly complex and time-consuming. Issues such as unbilled services, faulty meters, data entry errors, and abnormal consumption patterns can negatively impact revenue.

Service performance analysis helps identify gaps between total services and billed services. Consumption analysis provides insights into electricity usage trends across different regions. Accurate data analysis enables better planning and resource allocation. By analyzing service and consumption data, TGSPDCL can improve operational transparency and financial sustainability. The use of analytics tools helps transform raw data into actionable insights for effective decision-making.

3. OBJECTIVES OF THE PROJECT

The key objectives of this project are:

- To analyze total services vs billed services across divisions and areas.
- To evaluate billing efficiency (%) and identify low-performing regions.
- To study electricity unit consumption patterns area-wise and division-wise.
- To detect billing gaps and service performance issues using visual analytics.
- To compare performance across areas, divisions, and sections

4. SOURCE OF DATASET

Electricity service performance and consumption-related data published by the **Telangana State Southern Power Distribution Company Limited (TGSPDCL)** has been utilized. TGSPDCL is a government-owned electricity distribution company responsible for power supply in the southern districts of Telangana. As part of its operational reporting and transparency efforts, TGSPDCL periodically shares aggregated service and consumption data through the Telangana Open Data Portal.

Link: <https://data.telangana.gov.in/>

The key attributes included in the dataset are as follows:

- **Circle** – Represents the major administrative electricity distribution zone.
- **Division** – Sub-unit under a circle responsible for local distribution management.
- **Area / Section** – Smaller operational region handling service connections.
- **Total Services** – Total number of electricity service connections in the area.
- **Billed Services** – Number of services that were successfully billed.
- **Unbilled Services** – Services that were not billed due to technical or operational issues.

The dataset contains 71,628 records with 11 columns, representing power consumption, service counts, and billing details across various administrative levels (Circle, Division, Subdivision, Section, Area). It appears to be an operational and billing dataset for electricity services.

5. TOOLS AND TECHNOLOGIES USED

- **Power BI Desktop:** Power BI Desktop is a business intelligence and data visualization tool developed by Microsoft. It is used to connect to data sources, create data models, and build interactive reports and dashboards. In this project, Power BI Desktop is used to analyze TGSPDCL service performance and consumption data and to design visuals such as charts, tables, KPIs, and slicers for better decision-making.
- **Power Query:** Power Query Editor is a data transformation tool within Power BI used for data cleaning and preprocessing. It helps in importing, filtering, sorting, renaming columns, handling missing values, and creating calculated columns. In this project, Power Query Editor is used to prepare the dataset and calculate billing efficiency without using DAX, ensuring clean and reliable data for analysis.
- **Microsoft Excel / CSV files:** are used as the data source format. These files contain structured electricity service and consumption data that can be easily imported into Power BI

6. EXPLORATORY DATA ANALYSIS (EDA) PROCESS

Exploratory Data Analysis (EDA) is a critical phase in any data analytics project, as it helps in understanding the dataset before applying detailed analysis or visualization techniques. The primary objective of EDA is to examine the data structure, identify trends, detect inconsistencies, and ensure overall data quality. In the context of public utility organizations such as electricity distribution companies, accurate data analysis is extremely important because decisions based on incorrect or incomplete data can directly impact revenue, service reliability, and customer satisfaction.

In this project, EDA was carried out on TGSPDCL service performance and electricity consumption data obtained from structured operational records. The dataset contains information related to division-wise and area-wise service connections, total services, billed services, electricity units consumed, and performance indicators. Since the data represents large-scale electricity distribution operations across multiple regions, a systematic EDA process was necessary to ensure the data was accurate, consistent, and suitable for analysis.

6.1 Data Import and Initial Inspection

The TGSPDCL service and consumption dataset was imported into **Power BI Desktop** from structured CSV/Excel files. After loading the data, an initial inspection was carried out to understand the number of records, column names, and data types. The dataset mainly consisted of service-related metrics such as division, area, total services, billed services, and electricity units consumed. This step helped in gaining a high-level understanding of the dataset structure and identifying fields relevant for analysis.

6.2 Data Profiling and Quality Check

Power Query Editor's **column profiling tools** (Column Quality, Column Distribution, and Column Profile) were used to examine the quality of the data. This helped in identifying null values, zero values, and inconsistent entries in numeric columns. Special attention was given to service count and billing columns, as inaccuracies in these fields could affect billing efficiency calculations. Data profiling ensured that the dataset was reliable and ready for transformation.

6.3 Data Cleaning and Preparation

Data cleaning was performed entirely in **Power Query Editor**. Unnecessary columns were removed, and relevant columns were renamed to improve clarity and readability. Data types were corrected by converting service counts and consumption values into numeric formats. Text fields such as division and area names were standardized to avoid duplication and mismatch issues. This step ensured consistency across the dataset.

6.4 Handling Missing and Zero Values

During EDA, it was observed that some records contained zero or missing values in total services and billed services. To avoid incorrect calculations, conditional logic was applied in Power Query. Billing efficiency was calculated only when total services were greater than zero; otherwise, null values were assigned. This prevented division errors and improved the accuracy of performance analysis.

6.5 Creation of Derived Columns

As part of EDA, necessary derived columns were created using **Power Query transformations**. A key derived metric, **Billing Efficiency**, was calculated by dividing billed services by total services. This calculation was performed in Power Query instead of DAX to ensure backend-level data consistency. These derived columns played a vital role in performance comparison and visualization.

6.6 Aggregation and Grouping Analysis

The dataset was grouped by division and area to analyze service performance at different organizational levels. Aggregation operations such as sum and average were applied to service counts and consumption values. This grouping helped in understanding regional variations and identifying high-performing and low-performing divisions. These grouped results were later used directly in dashboard visuals.

6.7 Relationship Validation

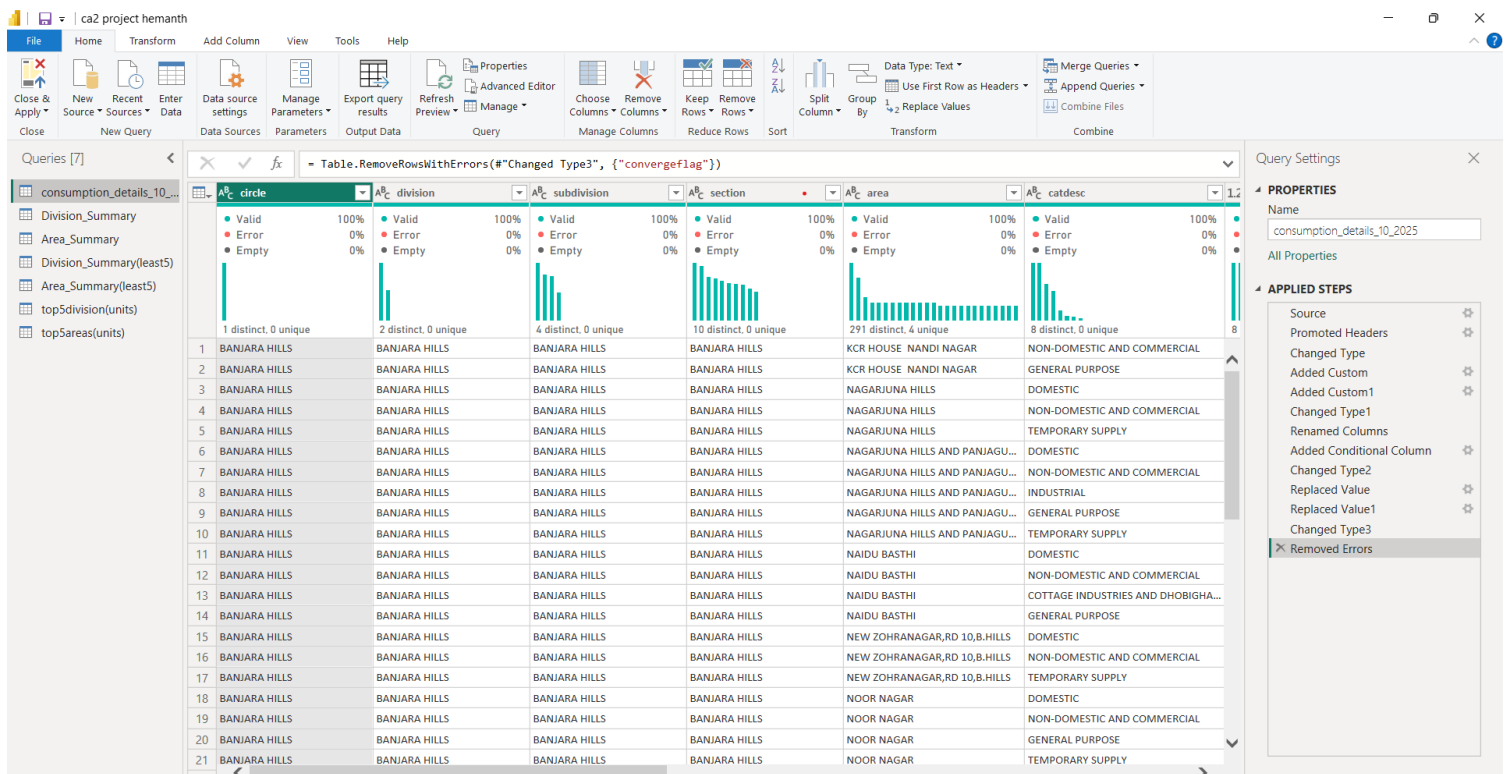
Where multiple tables were involved, relationships were verified to ensure accurate data interaction. Proper relationships allowed cross-filtering between visuals and ensured consistent results across different dashboard views. This step was important for enabling drill-down and slicer-based interactivity in the final dashboard.

6.8 Outcomes of Exploratory Data Analysis

The EDA process resulted in a **clean, structured, and reliable dataset** ready for visualization. Key outcomes include:

- Identification of data quality issues and their resolution
- Accurate calculation of billing efficiency
- Clear understanding of service and consumption patterns
- Prepared dataset suitable for interactive dashboards

EDA helped uncover early insights such as variations in billing efficiency and consumption across divisions. These outcomes directly supported the dashboard design and key findings of the project.



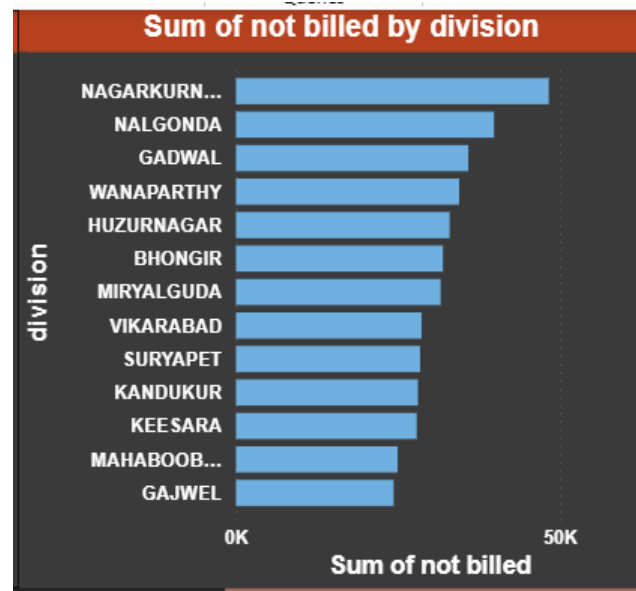
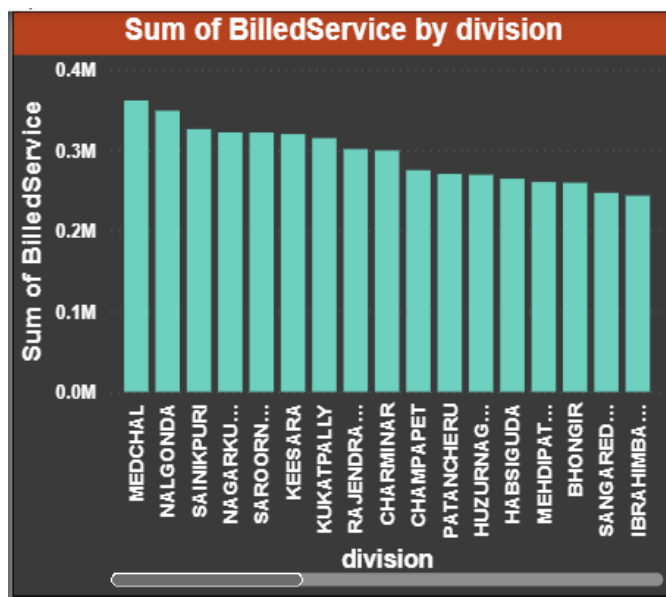
7. ANALYSIS ON DATASET

After completing the Exploratory Data Analysis (EDA), detailed analysis was carried out on the TGSPDCL dataset using Power BI Desktop. The analysis focuses on service performance, billing efficiency, and electricity consumption patterns across divisions and areas. The insights were derived through aggregation, comparison, and visualization techniques implemented in the Power BI dashboard.

7.1 Division-wise Service Performance Analysis

This analysis examines the total number of electricity services available in each division and compares them with the number of billed services. The purpose of this analysis is to understand how effectively services are being billed across different divisions.

By aggregating total services and billed services at the division level, divisions with a large gap between these two metrics were identified. Such gaps indicate possible operational issues like unbilled connections, meter faults, or data inconsistencies. High-performing divisions show a close match between total and billed services, indicating efficient service management.



7.2 Billing Efficiency Analysis

Billing efficiency is a key performance indicator in this project. It is calculated as the ratio of billed services to total services using Power Query conditional logic. This analysis helps identify divisions and areas where billing performance is low.

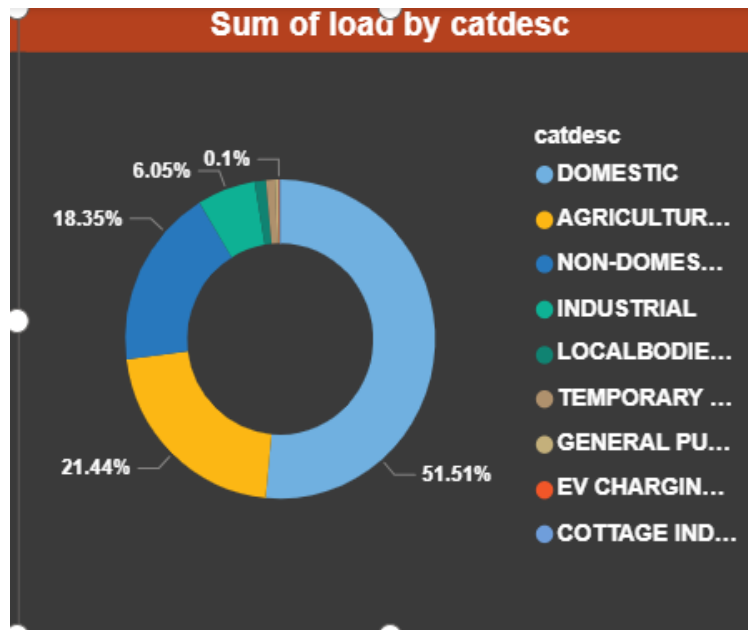
The dashboard visually highlights variations in billing efficiency across regions. Areas with low billing efficiency are critical zones that require immediate attention from management. Ranking visuals were used to compare top-performing and bottom-performing divisions based on billing efficiency.

10.85M Sum of BilledService	division	Sum of TotalService	Billing percentage at division	Sum of Billed Service	convergeflag
	▲				
	ASMANGADH	2,40,196.00	0.96	2,30,381.00	Good
	AZAMABAD	2,09,013.00	0.97	2,01,909.00	Good
	BANJARA HILLS	2,00,100.00	0.91	1,82,702.00	Good
	BEGUM BAZAR	1,84,254.00	0.92	1,69,807.00	Good
	BHONGIR	2,91,546.00	0.89	2,59,597.00	Moderate
	BOWENPALLY	1,49,306.00	0.96	1,43,363.00	Good
	CHAMPAPET	2,82,229.00	0.98	2,75,200.00	Good
	CHARMINAR	3,19,758.00	0.94	2,99,677.00	Good
	CHOUTUPPAL	1,62,026.00	0.91	1,47,655.00	Good
	DEVARAKOND A	1,79,592.00	0.90	1,61,399.00	Moderate
	GACHIBOWLI	2,38,176.00	0.95	2,25,923.00	Good
	GADWAL	2,42,619.00	0.85	2,06,759.00	Moderate
	GAJWEL	2,05,737.00	0.88	1,81,405.00	Moderate
	Total	1,17,94,044.00		1,08,52,657.00	
11.79M Sum of Totalservice					
941K Sum of not billed					

7.3 Electricity Consumption Analysis

This analysis focuses on electricity unit consumption across different divisions and areas. Total units consumed were aggregated to understand consumption behavior and demand distribution.

High-consumption areas were identified and compared against billing performance to check whether higher consumption is being accurately billed. This analysis helps detect abnormal consumption patterns and supports demand planning and load management decisions.



7.4 Area-wise Comparative Analysis

Area-wise analysis was performed to provide a more granular view of service performance and consumption. Using slicers and filters, users can select specific divisions and analyze individual areas within them.

This comparison helps identify specific sections that are underperforming even if the overall division performance appears satisfactory. Such micro-level insights are useful for targeted corrective actions.

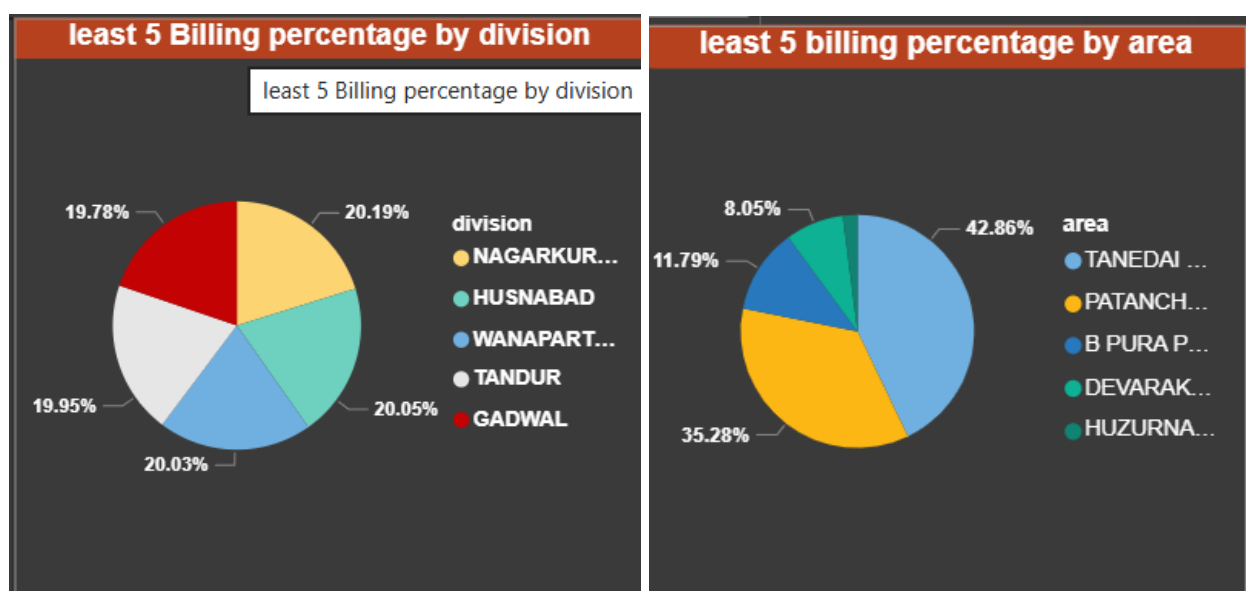
area	Sum of Totalservice	Sum of Billing percentage at area	Sum of BilledService	converge flag
BALAJI NAGAR	33,827.00	1.92	32,878.00	Good
VIKARABAD TOWN	21,514.00	0.92	19,715.00	Good
INDIRA NAGAR	20,976.00	0.96	20,174.00	Good
MEDAK TOWN	19,949.00	1.95	19,426.00	Good
GANESH NAGAR	19,745.00	1.95	18,934.00	Good
HANUMAN NAGAR	19,479.00	0.96	18,697.00	Good
SHANTHI NAGAR	19,293.00	0.96	18,449.00	Good
DEVARAKONDA	15,736.00	1.97	15,203.00	Good
MOOSAPET	15,218.00	0.94	14,353.00	Good
KONDAPUR	14,741.00	0.90	13,204.00	Moderate
MARUTHI NAGAR	14,634.00	0.98	14,335.00	Good
PARGI	13,344.00	0.92	12,242.00	Good
Total	1,17,94,044.00	13,778.68	1,08,52,657.00	

circle, area
☒ BANJARA HILLS
☒ CYBERCITY
☒ GADWAL
☒ HABSIGUDA
☒ HYDERABAD CENT...
☒ HYDERABAD SOUTH
☒ MAHABOONNAGAR
☒ MEDAK
☒ MEDCHAL
☒ NAGARKURNOOL

7.5 Identification of Low-Performing Zones

By combining service count, billing efficiency, and consumption metrics, low-performing zones were identified. These zones typically show low billing efficiency, high service gaps, or abnormal consumption values.

This analysis supports management in prioritizing inspections, meter audits, and operational improvements. The dashboard makes it easy to visually detect these zones through color-coded and ranked visuals.



7.6 Overall Analytical Outcomes

The dataset analysis provided clear insights into TGSPDCL's operational performance. It revealed variations in billing efficiency, consumption patterns, and service management across divisions and areas. The analysis validated the effectiveness of using Power BI for public utility data analysis and supported data-driven decision-making.

TGSPDCL Service Performance Dashboard

10.85M

Sum of BilledService

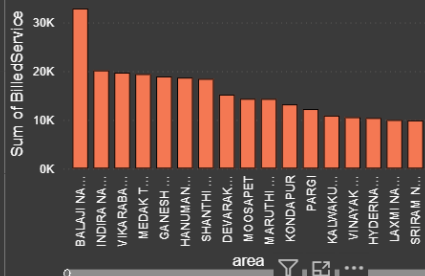
11.79M

Sum of Totalservice

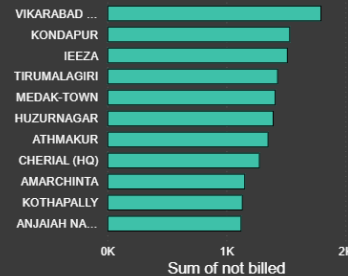
941K

Sum of not billed

Sum of BilledService by area



Sum of not billed by area

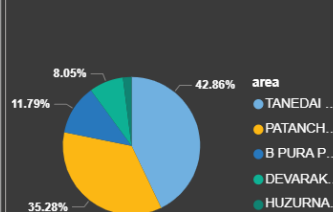


circle, area

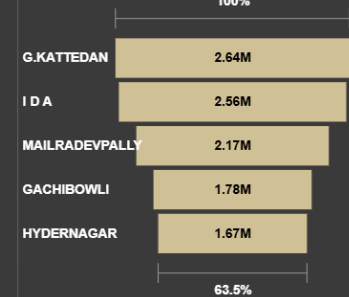
- ☒ BANJARA HILLS
- ☒ CYBERCITY
- ☒ GADWAL
- ☒ HABSIGUDA
- ☒ HYDERABAD CENT...
- ☒ HYDERABAD SOUTH
- ☒ MAHABOONNAGAR
- ☒ MEDAK
- ☒ MEDCHAL
- ☒ NAGARKURNOOL

area	Sum of Totalservice	Sum of Billing percentage at area	Sum of BilledService	converge flag
BALAJI NAGAR	33,827.00	1.92	32,878.00	Good
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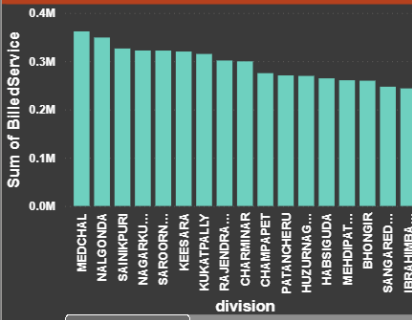
least 5 billing percentage by area



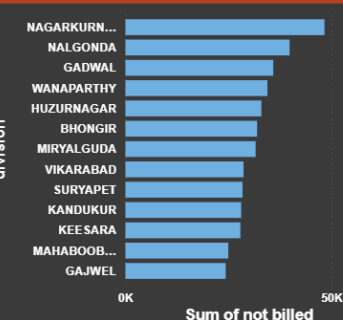
Top 5 Highest units taken by area



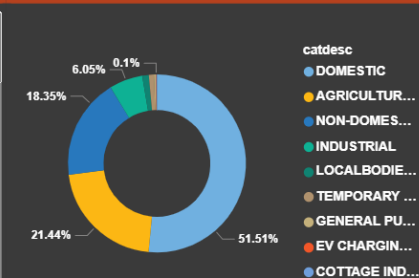
Sum of BilledService by division



Sum of not billed by division

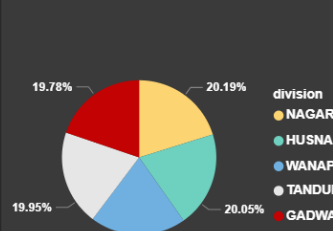


Sum of load by catdesc

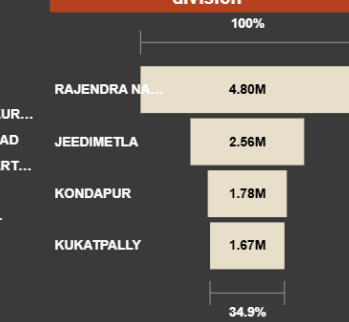


division	Sum of Totalservice	Billing percentage at division	Sum of BilledService	convergeflag
ASMANGADH	2,40,196.00	0.96	2,30,381.00	Good
AZAMABAD	2,09,013.00	0.97	2,01,909.00	Good
BANJARA HILLS	2,00,100.00	0.91	1,82,702.00	Good
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GAJWEL	2,05,737.00	0.88	1,81,405.00	Moderate
Total	1,17,94,044.00		1,08,52,657.00	

least 5 Billing percentage by division



Top 5 Highest units taken by division



8. DASHBOARD DESIGN AND INTERACTIVITY

The dashboard in this project was designed using Power BI Desktop with a focus on clarity, usability, and performance monitoring. The design emphasizes service performance, billing efficiency, and electricity consumption analysis. Simple and meaningful visuals were selected to ensure easy interpretation by users at different management levels.

8.1 Dashboard Layout Design

A single-page dashboard layout was used to present all key performance indicators in one view. The layout follows a top-to-bottom and left-to-right structure, allowing users to understand overall performance first and then move to detailed analysis. Proper spacing and alignment were maintained to avoid visual clutter and improve readability.

8.2 Key Performance Indicator (KPI) Cards

KPI cards were used to display high-level metrics such as **Total Services**, **Billed Services**, **Billing Efficiency**, and **Total Units Consumed**. These KPIs provide a quick summary of system performance and help users instantly assess the current status of service and billing operations.

8.3 Visual Charts Used

Bar charts and column charts were used to represent **division-wise service performance**, **billing efficiency**, and **electricity consumption**. These visuals allow easy comparison across divisions and help identify top-performing and low-performing regions. Tables were used to present detailed area-level data for deeper analysis.

8.4 Use of Slicers for Filtering

Slicers were implemented to filter data by **division and area**. These slicers allow users to interact with the dashboard and focus on specific regions. When a slicer is applied, all related visuals update dynamically, enabling targeted analysis without modifying the underlying data.

8.5 Drill-Down and Cross-Filtering

Drill-down functionality was used to move from **division-level analysis to area-level details** within the same visual. Cross-filtering enables interaction between visuals, where selecting a value in one chart automatically filters related visuals. This enhances data exploration and insight discovery.

8.6 Interactivity Outcomes

The interactive features of the dashboard make it easy for users to explore service performance and consumption patterns. Decision-makers can quickly identify problem areas, compare regions, and analyze performance trends. The dashboard provides an effective, user-friendly platform for data-driven decision-making.

9. KEY INSIGHTS AND FINDINGS

- Billing efficiency varies significantly across different TGSPDCL divisions, indicating uneven service and billing performance.
- Several divisions have a noticeable gap between total services and billed services, highlighting possible unbilled connections or operational issues.
- High electricity consumption is concentrated in specific divisions, reflecting regional demand differences.
- Some high-consumption areas show lower billing efficiency, suggesting the need for improved meter reading and billing accuracy.
- Area-level analysis helped identify low-performing sections within otherwise better-performing divisions.
- Ranking visuals clearly highlight top-performing and bottom-performing divisions based on billing efficiency.
- Interactive slicers enabled focused analysis of specific divisions and areas, improving insight accuracy.
- Cross-filtering between visuals made it easier to detect relationships between service count, billing efficiency, and consumption.
- The dashboard effectively supports identification of priority zones for corrective action.
- Overall analysis demonstrates the usefulness of Power BI for monitoring service performance and consumption in electricity distribution.

10. CONCLUSION

This project successfully demonstrates the application of Power BI for analyzing TGSPDCL service performance and electricity consumption data. By using Power Query for data cleaning, transformation, and calculation, the project ensured accurate and reliable analysis without relying on complex DAX formulas. The developed dashboard provides a clear and interactive view of key metrics such as total services, billed services, billing efficiency, and electricity units consumed.

The analysis helped identify variations in billing efficiency across divisions and areas, highlighting operational gaps and potential revenue leakage zones. Consumption analysis offered valuable insights into regional electricity demand patterns, supporting better planning and resource allocation. Area-level analysis further enabled detailed monitoring and identification of underperforming sections.

Overall, the project proves that interactive dashboards can significantly improve transparency, monitoring, and decision-making in the power distribution sector. The insights derived from this analysis can assist TGSPDCL management in taking corrective actions, improving billing performance, and enhancing overall service efficiency. This project also highlights the effectiveness of Power BI as a practical and powerful business intelligence tool for public utility data analysis.

11. FUTURE SCOPE

The current project provides valuable insights into TGSPDCL service performance and electricity consumption; however, there are several opportunities to enhance and extend the scope of this work in the future. One major enhancement could be the integration of real-time or near real-time data, allowing continuous monitoring of service performance and billing efficiency. This would enable faster identification of issues and quicker corrective actions.

Advanced analytics techniques such as predictive analysis can be incorporated to forecast future electricity consumption and identify potential billing risks. The use of DAX functions can further improve analytical depth by enabling complex calculations and time-based comparisons. Additionally, the dashboard can be extended to include historical trend analysis for better performance evaluation over time.

The project can also be expanded by integrating customer complaint data, outage information, and meter status data to provide a more comprehensive view of operational performance. Publishing the dashboard using Power BI Service would enable automated data refresh, role-based access, and easier sharing among stakeholders. Incorporating AI-driven insights and anomaly detection features could further help identify unusual consumption or billing patterns.

Overall, expanding the scope of this project would strengthen its usefulness as a decision-support system and support long-term planning, operational efficiency, and service improvement for TGSPDCL.

12. REFERENCES

1. Telangana State Southern Power Distribution Company Limited (TGSPDCL)– Official reports and service performance data
Link: <https://data.telangana.gov.in/>
2. Microsoft Power BI Documentation – Official guide for Power BI Desktop and Power Query Editor.

13. LINKEDIN

Link: https://www.linkedin.com/posts/hemanth-ponamala_powerbi-dataanalytics-businessintelligence-activity-7406028359941423104-ogfA?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEg017wB56CZwyQhgbL7Euo2LxHtZiM6Tq4

Screenshot:

